

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: To analyze sensor data in Flutter App.	
Experiment No:- 08	Date:- 06-12-2024	Enrollment No:- 92200133030	

Objective:- To analyze sensor data in Flutter App.

Code :-

➤ **lib/main.dart**

```
import 'package:flutter/material.dart';
import "package:sensors_plus/sensors_plus.dart";
import 'dart:async';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Sensor Data Analyzer',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const SensorPage(),
    );
  }
}

class SensorPage extends StatefulWidget {
  const SensorPage({super.key});

  @override
  // ignore: library_private_types_in_public_api
  _SensorPageState createState() => _SensorPageState();
}

class _SensorPageState extends State<SensorPage> {
  // Accelerometer sensor values
  double _accelerometerX = 0.0;
  double _accelerometerY = 0.0;
  double _accelerometerZ = 0.0;

  // Gyroscope sensor values
  double _gyroscopeX = 0.0;
  double _gyroscopeY = 0.0;
  double _gyroscopeZ = 0.0;
```

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```
// Magnetometer sensor values
double _magnetometerX = 0.0;
double _magnetometerY = 0.0;
double _magnetometerZ = 0.0;

// Stream subscriptions
StreamSubscription<AccelerometerEvent>? _accelerometerSubscription;
StreamSubscription<GyroscopeEvent>? _gyroscopeSubscription;
StreamSubscription<MagnetometerEvent>? _magnetometerSubscription;

@override
void initState() {
  super.initState();

  // Initialize Accelerometer listener
  _accelerometerSubscription =
    // ignore: deprecated_member_use
    accelerometerEvents.listen((AccelerometerEvent event) {
      setState(() {
        _accelerometerX = event.x;
        _accelerometerY = event.y;
        _accelerometerZ = event.z;
      });
    });

  // Initialize Gyroscope listener
  // ignore: deprecated_member_use
  _gyroscopeSubscription = gyroscopeEvents.listen((GyroscopeEvent event) {
    setState(() {
      _gyroscopeX = event.x;
      _gyroscopeY = event.y;
      _gyroscopeZ = event.z;
    });
  });

  // Initialize Magnetometer listener
  _magnetometerSubscription =
    // ignore: deprecated_member_use
    magnetometerEvents.listen((MagnetometerEvent event) {
      setState(() {
        _magnetometerX = event.x;
        _magnetometerY = event.y;
        _magnetometerZ = event.z;
      });
    });
}

@override
void dispose() {
  // Cancel subscriptions to avoid memory leaks
  _accelerometerSubscription?.cancel();
  _gyroscopeSubscription?.cancel();
}
```

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```

    _magnetometerSubscription?.cancel();
    super.dispose();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Sensor Data Analyzer'),
    ),
    body: Padding(
      padding: const EdgeInsets.all(16.0),
      child: ListView(
        children: [
          const Text(
            'Accelerometer:',
            style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
          ),
          Text('X: ${_accelerometerX.toStringAsFixed(2)}'),
          Text('Y: ${_accelerometerY.toStringAsFixed(2)}'),
          Text('Z: ${_accelerometerZ.toStringAsFixed(2)}'),
          const SizedBox(height: 20),
          const Text(
            'Gyroscope:',
            style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
          ),
          Text('X: ${_gyroscopeX.toStringAsFixed(2)}'),
          Text('Y: ${_gyroscopeY.toStringAsFixed(2)}'),
          Text('Z: ${_gyroscopeZ.toStringAsFixed(2)}'),
          const SizedBox(height: 20),
          const Text(
            'Magnetometer:',
            style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
          ),
          Text('X: ${_magnetometerX.toStringAsFixed(2)}'),
          Text('Y: ${_magnetometerY.toStringAsFixed(2)}'),
          Text('Z: ${_magnetometerZ.toStringAsFixed(2)}'),
        ],
      ),
    ),
  );
}

```

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Output:-

